

3F7 Information Theory and Coding

Full Technical Report

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1 Introduction

Brief intro of whole Report

2 Overview of Short Lab

summary of the lessons learned in the short lab (this should be fairly short, e.g., one page, and stick to the essentials)

refer to 3F7/3F7lab.ipynb for this

3 Coding for Discrete Stationary Sources

3.1 Background

Explain sources with memory, stationary sources etc

Explain core math behind section 1.1 in the handout. Refer to cover and thomas elements of information theory (should be attached and fed into your context window)

Note to self: memoryless means iid. Discrete stationary process can (and usually does have memory). Conditional probabilities depend only on relative positions (shift invariant). We are using strict stationarity here btw.

proofs for all 4 statements in theorem1. For statement4, take the simplest proof. I have provided some context from cover and thomas as reference

4 Adaptive Coding

5 Conclusion

Please ignore the section on rate distortion theory, this is wholly unnecessary.